Projects for Machine Learning

A.A. 2025/26

This document describes the projects for Machine Learning during the academic year 2025/2026 (hence, this document is valid only for the academic year 2025/2026).
Projects are mandatory. The project discussion will contribute to two thirds (2/3) of the total grade.

Instructions

There will be four projects, designed by professionals from two different companies: Whitehall Reply and Alkemy.

You will work on one of these projects in teams of **at most three students** (this is a strict constraint!). You are responsible for forming your own teams, and each team must appoint a captain, who will act as the main contact person for the project.

After reviewing the projects available on **my.luiss.it**, the team captain must submit the project preference by filling out the designated form no later than **March 5 at 23:59 CET.**

Your email address will be used to establish contact with the company running the project, so please make sure that (a) all email addresses are correct, and (b) you agree to share your email address with the company.

We will try our best to assign each team a project according to the stated preference. However, it is also important that the teams spread across the different projects. Thus, we will try to balance different projects and, generally, we will apply a first come first served approach. If you do not fill in the form by the given deadline, you will be assigned to a project and to a team by the TAs.

Mid check and final pitch

There will be a checkpoint in the middle of the project held online through the webex rooms of the TAs in April 2026.

There will be a project final pitch in May 2026 held online through the webex rooms of the TAs. For the final and specific dates, please refer to the course page on **my.luiss**.

When and what to submit

You must submit your project by **May 1st, 2025, at 11:59 PM.**

The project submission must be coordinated with the company running the project, i.e., Whitehall Reply or Alkemy.

Each group must also **submit via** form the URL of the **GitHub repository** containing also the **presentation** used for the final pitch. The repository's name must end with the student id of the “captain”.

The repository must contain:

1. A “README.md” file with the following information:
    - Title and Team members
    - [Section 1] Introduction – Briefly describe your project.
    - [Section 2] Methods – Describe your proposed ideas (e.g., features, design choices, algorithm(s), training overview, etc.) and your environment so that:
        - A reader can understand why you made your design decisions and the reasons behind any other choice related to the project.
        - A reader should be able to recreate your environment (e.g., conda list, conda envexport, etc.)
        - It may help to include a figure illustrating your ideas, e.g., a flowchart illustrating the steps in your machine learning system(s)
    - [Section 3] Experimental Design – Describe any experiments you conducted to demonstrate/validate the target contribution(s) of your project; indicate the following for each experiment:
        - The main purpose: 1-2 sentence high-level explanation
        - Baseline(s): describe the method(s) that you used to compare your work to.
        - Evaluation Metrics(s): which ones did you use and why?
    - [Section 4] Results – Describe the following:
        - Main finding(s): report your results and what you might conclude from your work.
        - Include at least one placeholder figure and/or table for communicating your findings.

  - All the figures containing results should be generated from the code.
- [Section 5] Conclusions – List some concluding remarks. In particular:
  - Summarize in one paragraph the take-away point from your work.
  - Include one paragraph to explain what questions may not be fully answered by your work as well as natural next steps for this direction of future work.

2. A single notebook called “main.ipynb” with ALL the code used for the project. The

   notebook must have the following characteristics:
   - Text and code cells must alternate from start to finish. The text cell above must describe the contents of the code below and its output so that a reader can easily follow up on your implementation. In particular:
     - You must explain what you will do and why you chose to do so.
     - You must explain the outputs of the cell (if any) with particular attention to describing figures such that a reader already knows what he is going to see.
3. An additional folder named “images” contains the figures displayed in the “README.md”.
4. Each group must also submit any additional deliverables requested by the company running the project.